Jithendra Puppala

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Summary

MS CS at NYU with 2+ years as a Data Scientist at Jio Platforms, shipping production ML to 150K+ IoT devices ($10M+ annual savings). Built kernel search and on-chip validation for AI-written NKI kernels on AWS Trainium2, and found and fixed a precision bug in AWS's NKI tutorial. CV/ML intern at Applied Materials, automating wafer-inspection recipe tuning with Bayesian optimization. Kaggle: 1st in computer vision (50+ teams), 3rd in tabular ML (100+ teams).

Education

New York University — New York, NY
Master of Science in Computer Science | GPA: 3.83 — Expected: 05/2027

- Coursework: Deep Learning (Prof. **Yann LeCun**), Machine Learning, Computer Vision, Big Data
- **Work Authorization:** US work-eligible via CPT (internship) and STEM OPT (3 years post-graduation)

National Institute of Technology Karnataka — Surathkal, India
Bachelor of Technology in Computer Science and Engineering — 05/2023

Skills

- **Accelerators & Kernels**: AWS Trainium2, Neuron SDK, NKI, CUDA, bf16/fp32 Numerics, On-Device Benchmarking & Profiling
- **Languages & Core ML**: Python, SQL, PyTorch, TensorFlow, Scikit-learn, XGBoost, LightGBM, Hugging Face, LangChain
- **Deep Learning**: Vision Transformers (ViT, Swin, EVA02), CNNs (ConvNeXt), LLMs/Transformers, CUDA, Mixed Precision, Flash Attention, Natural Language Processing
- **Computer Vision**: OpenCV, YOLO, Object Detection & Tracking, Image Classification, Multi-View Fusion
- **MLOps & Cloud**: Docker, FastAPI, REST APIs, AWS (EC2, S3), GCP, Azure, PySpark, MLflow, CI/CD, Git, Linux

Work Experience

Applied Materials — Santa Clara, CA
Computer Vision & Machine Learning Intern — 06/2026 – 08/2026

- Reduced **missed defects by 67%** (held-out coverage 95.1% to 98.4%) at a fixed false-alarm budget, or **false alarms 5.4×** at matched recall, via **bilevel Bayesian optimization** of wafer-inspection recipes
- Discovered an image feature separating defects from nuisance at **AUC 0.956** vs. the engine's best 0.693 (200/200 bootstrap wins); showed Bayesian search recovers the production recipe in a median of **13 evaluations vs. 103**
- Caught a coverage metric reporting a **false 100%** (85% of on-defect detections were on the wrong die), fixed it, and shipped **1,702 automated tests** and an offline React + FastAPI dashboard

New York University – Office of Faculty Affairs — New York, NY
Lead Developer – Data Science & AI — 01/2026 – Present

- Built **RAG** ticketing prototype (LangChain, ChromaDB, sentence-transformers) for faculty document Q&A at **94%** retrieval accuracy, scaling to auto-resolve **900+ annual requests** across **15+ administrative workflows**

Jio Platforms Limited — Bengaluru, India
Data Scientist — 06/2023 – 07/2025

- Owned end-to-end ML classification pipeline serving real-time predictions to **150K+ IoT devices** via FastAPI microservices, improving **precision from 68% to 91%** and **recall from 76% to 95%**
- Built ARIMA-based anomaly detection that **reduced alert latency from 12 hours to 4 hours** (67% faster), improving field response success rates by **28%** and driving **$10M+** in annual savings
- Re-architected ML retraining infrastructure: model retraining from **18 hours to 2 hours** (89% reduction), data processing (ETL) from **3 hours to 7 minutes** (96% reduction)

Projects

CHIPBOOST: AI-Written NKI Kernels on AWS Trainium2 (Annapurna Labs Hackathon; merged) — 10/2026

- Built kernel search and on-chip validation for an agent where Qwen3-8B writes NKI matmul kernels under a red-teamed referee; reached a verified **1.52×** speedup, and **1.375×** over AWS's default via block-size tuning
- Found and fixed a precision **bug in AWS's official NKI matmul tutorial** (**19.4 → 0.5** bf16 ulps); filed **issue #1422** with **fix PR #1423**

Multi-View Vision Transformers for Geospatial Prediction *(1st Place, Kaggle, 50+ Teams)* — 11/2025 – 01/2026

- Developed multi-view attention architecture fusing 4 street-view images for 50-state classification + GPS regression, fine-tuning 7 vision backbones (ViT-CLIP, EVA02, Swin, BEiT, ConvNeXt)
- Achieved **95.7% score** via weighted ensemble with ablation-informed model selection and Test-Time Augmentation (TTA)
- Engineered GPU-optimized pipeline with **tensor caching** and **on-GPU** MixUp & CutMix, cutting A100 epoch time **40%**

Leadership & Achievements

- **CV Engineer, NYU ARC Robotics (RoboMaster)** – perception & object detection for competitive robotics
- **Harvard CELP Finalist (Top 114 of 10,000+)** | **JEE Advanced Rank: 6111** – Top **0.5% of 1.15M** candidates